

Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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Abstract

Households contribute 13% of U.S. carbon emissions, yet most people rely on biased decision making rather than tailored choices. Multi-Attribute Value Theory (MAVT) offers a rigorous method of Multi-Criteria Decision Analysis (MCDA) to reduce emissions, but building MCDAs daily requires expertise that limits household adoption. This study evaluated whether LLMs can bridge that gap by comparing three architectures: Pure Prompting, Retrieval-Augmented Generation (RAG-Enhanced) Scoring, and a Hybrid of LLM parameter extraction with calculation. Ground truth calculators were built to benchmark MAVT scores across HVAC, appliance, and shower decisions. *[Quantitative claims in this paragraph are BLOCKED pending corrected multi-model results; see audit log.]*

Keywords: Large language models, Multi-criteria decision analysis, Multi-Attribute Value Theory, Household energy, Decision support systems, Retrieval-Augmented Generation, Sustainability

1. Introduction

The average U.S. household emits roughly 8,744 pounds of CO₂e and consumes about 10,791 kWh of electricity each year [1, 2]. When transportation, food, and goods are folded in, per-capita footprints rise to about 17.9 tons of CO₂e and 81,767 kWh of total annual energy [3, 4]. The U.S. Department of Energy estimates that households can save \$200–\$400 annually by adopting ENERGY STAR practices [5, 6].

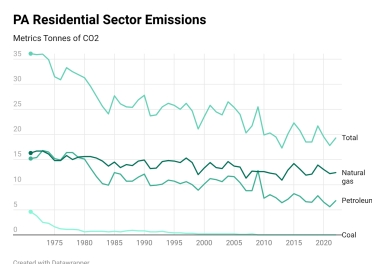


Figure 1: Residential emissions over the past 50 years (EIA, 2022) [7, 8].

A persistent gap between intent and behavior keeps many households from realizing those savings [9]. Tools that combine optimization, behavioral insight, and decision support can close part of that gap, but only if they are usable without specialized training.

Multi-criteria decision analysis (MCDA), life-cycle assessment (LCA), and Pareto-based prediction methods have been used to track and reduce environmental impact at building and organizational scales [10, 11]. Engineering frameworks of this kind can optimize complex decisions, but their construction is time-consuming: signal-processing

pipelines such as the Hilbert–Huang transform are not rapid, and constructing the decision matrix itself is a recognized bottleneck [12]. Huang and Burgherr (2024) addressed part of this by building a streamlined MCDA calculator that compresses several accepted methods into a single workflow [13]. Even with their simplification, the user still needs enough familiarity with MCDA construction to frame the alternatives, criteria, and weights, which keeps the tool out of routine household use.

Residential energy consumption is not uniformly distributed across end uses. Space heating accounts for roughly 43% of residential energy use, followed by appliances at 13%, water heating at 12%, and air conditioning at 8% [7]. Dietz et al. (2009) showed that a coordinated set of household behavioral changes could cut U.S. emissions by approximately 123 million metric tonnes of carbon (MtC) per year while preserving occupant satisfaction [14]. Behavioral plasticity, the likelihood that a household adopts a given action given financial and social context, varies substantially across actions and is the limiting factor for that wedge.

Figure 2 reproduces the Dietz et al. display of behavioral plasticity, net carbon effect, and the Reasonably Achievable Emission Reduction (RAER) per action.

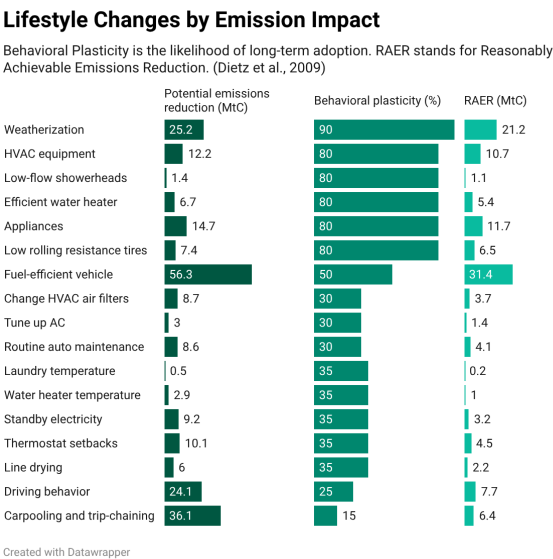


Figure 2: Household behavior changes and their emissions impact (after Dietz et al., 2009 [14]).

A practical obstacle for MCDA-style tools at the household level is the workflow tax they impose on users. Even when the benefits are clear, switching from a routine choice to a model-mediated decision adds time and cognitive cost, both of which are documented deterrents to adoption [15, 16, 17]. Most household decisions in this space, including thermostat setpoints, appliance run times, and shower length, are heuristic rather than analytical.

Heuristic decision-making is efficient for everyday choices but degrades in complex or high-stakes settings. Under time pressure, people default to satisficing, choosing the first option that clears a minimum threshold, or to dominant prior behaviors, even when better alternatives exist [18]. When time pressure combines with information overload, judgment error rates rise meaningfully [19, 20]. Household energy decisions typically occur in exactly this regime: short deliberation windows, competing demands, and incomplete information. Tools that require sustained attention or structured inputs are unlikely to influence behavior under those conditions.

AI is increasingly being used to build or operate decision frameworks in domains that share the household setting’s combination of variability and judgment. Generative AI now appears in operations and supply-chain decision pipelines [21], in marketing strategy [22], and in clinical decision support [23]. Its capacity to process heterogeneous, large-scale data suggests that it can scale down to individual-level decision support as well.

The harder problem is operational. Constructing an MCDA framework for a specific decision normally requires methodological knowledge, including the choice of weighting scheme, value function form, and normalization, alongside rubric design and domain familiarity [12, 13]. Existing simplified calculators reduce but do not eliminate this

barrier. A large language model that can convert a natural-language description of a household situation into the alternatives, criteria, and contextual parameters that an MCDA needs would, in principle, remove the methodological barrier altogether.

The plausibility of this approach is reinforced by recent work outside the energy domain. Svoboda and Lande (2024) used GPT-4 to construct an analytic hierarchy process (AHP) decision model for cybersecurity, reporting a consistency ratio below the conventional 0.10 acceptability threshold and noting that earlier models such as GPT-3 were unreliable [24]. That result indicates that current-generation LLMs can produce coherent MCDA structures, but it leaves open the separate question of whether they can produce accurate quantitative scores against a physics-based ground truth.

Modern LLMs also carry general knowledge about the energy use and carbon intensity of common products and behaviors, and many systems can retrieve additional information at inference time. They have not, however, been rigorously evaluated for individual-level household decisions against deterministic ground truth, and

2. Methodology

2.1. Decision Types and Scenario Corpus

Three specific decision types were chosen due to their high plasticity and RAER, as well as being timely, everyday decisions that may change constantly. These were HVAC-related decisions, which are about temperature control, Appliance-related decisions, which are about the time of day (TOD) to use appliances, and Shower-related decisions, which are about how long to shower for. Each of the 3 architectures were tested on a 265-scenario corpus that was divided into 130 HVAC, 80 Appliance, and 55 Shower decisions. They were compared to a calculated, decision-type specific dataset from a calculator. This calculator also ran on an additional 80 RAG scenarios, split into 35 HVAC, 30 Appliance, and 15 Shower for a retrieval augmented generation (RAG) database.

[TO UPDATE NUMBERS] The scenario corpus has two disjoint pools. The first is the test set: 181 scenarios drawn from the larger domain corpora (128 HVAC, 80 Appliance, 55 Shower) and stratified into 91 HVAC, 50 Appliance, and 40 Shower test cases. The second is the RAG corpus: 231 additional scenarios (111 HVAC, 90 Appliance, 30 Shower) that populate the retrieval index used by the RAG-Enhanced architecture. Each scenario presents three alternatives. Test-set scenarios are scored independently by each of the three architectures and by the ground-truth calculator; RAG-set scenarios are only used to seed the retrieval index and never appear in evaluation [25].

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The HVAC decision type was the most complex. The tables below detail the parameters given to architectures vs. calculators.

The calculator receives specific values that must be professionally graded for households, and thus are unreasonable to be requested of homeowners. The architectures receive generalized version (e.g. insulation, which is a general estimate of poor/average/good rather than a specific R-value) or do not receive counterparts to some parameters at all (e.g. occupancy context, which the architecture instead has to infer from the question text). These parameters are chosen as they are things a household member would actually know [26].

To generalize parameters, we identified acceptable ranges for each qualitative level from published standards and mapped each scenario to a representative value within that range. Insulation tiers map to standard U.S. residential wall R-values: Poor to R-11 (minimal 2×4 cavity fill, below current code), Medium to R-13 (standard 2×4 batt), and Good to R-19 (2×6 framing), consistent with code tables for wood-frame assemblies and ENERGY STAR guidance [27, 28]. For the Flow Rate parameter, if the scenario specified standard, the corresponding GPM assigned to the calculator was 2.5 GPM (the federal maximum under the Energy Policy Act of 1992) [29]; if `low_flow`, the value was 1.5 GPM (EPA WaterSense certified) [29]. Table 4 summarizes the generalized parameters across all three decision types.

Table 1: HVAC parameters: Architectures vs. calculator

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Home Size (sq ft)	✓	✓
Insulation	✓	—
R-Value	—	✓
Household Size	✓	✓
Housing Type	✓	✓
House Age ^a	✓	✓
Utility Budget	✓	✓
SEER Rating	—	✓
Occupancy Context	—	✓
Indoor Temp (from alternative)	—	✓

^aLLMs receive a building-age range label (e.g., “20+ Years”); the calculator uses HVAC equipment age in exact years — distinct concepts.

Table 2: Appliance parameters: Architectures vs. calculator.

Parameter	LLMs	Calc.
Appliance Type	✓	✓
kWh per Cycle	—	✓
Cycle Duration (min)	✓	✓
Preferred Run Time	✓	—
TOU Period	✓	✓
Baseline Time	✓	—
Efficiency Rating	✓	✓
Household Size	✓	✓
Utility Rate	✓	✓
Monthly Cycles Assumption	—	✓
Utility Budget	✓	✓

The two sets overlap on environmental conditions, however so that both pipelines are reasoning about the same physical scenario. The LLM never sees the engineering parameters directly. For the Hybrid architecture, which needs to use those parameters within its calculations, it infers structured estimates of them from the description before the calculator is invoked. Further detail on choosing these specific parameters and their effects is available in section 2.2.

2.2. MAVT Framework Design

The Multi-Attribute Value Theorem (MAVT) allows alternatives to be compared on different criterion based on the range of values that occurs for each criterion. In our study, we chose the following weights for our 4 weights: Environmental Impact (35%), Energy Cost (30%), Comfort (20%), and Practicality (15%). The weights are held constant across all decision types, all architectures, and the ground-truth calculator. We did this so that any difference in score is because of the inputs and value function, not because of difference in the way each criterion is perceived for the three decision types. Normally, MAVT additive form (Eq. 1) requires preferential independence among criteria [33, 34]. This means that criteria should be independent, meaning that the value (good or bad) of one criterion should have no effect on the value of another criterion.

$$s_j = \sum_{i=1}^4 w_i \cdot v_i(x_{ij}) \quad (1)$$

In the case of our criteria, the energy cost and the environmental impact criterion are not completely independent, as they both stem from the kWh usage of a certain activity. The spearman’s rho is *[value pending corrected metrics]*

Table 3: Shower parameters: Architectures vs. calculator.

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Inlet Water Temp	—	✓
Heater Setpoint	✓	✓
Flow Rate (standard/low_flow)	✓	—
GPM	—	✓
Duration (min)	✓	✓
Household Size	✓	✓
Water Heater Type	✓	✓
Tank Size	—	✓
Utility Budget	✓	✓
Occupancy Frequency	✓	—

Table 4: Parameters generalized from engineering values to homeowner-accessible labels. LLMs receive the label; calculators receive the numeric value. R-values in $\text{h}\cdot\text{ft}^2 \cdot ^\circ\text{F}/\text{BTU}$; GPM dataset range 1.5–4.0; kWh/cycle ranges span the full dataset.

Parameter	LLM Label	Calc. Value	Source
<i>HVAC</i>			
Insulation	Poor	R-11	[27, 28]
	Medium	R-13	
	Good	R-19	
<i>Shower</i>			
Flow Rate	low_flow	1.5 GPM	[29]
	standard	2.5 GPM	
<i>Appliance</i>			
Appliance Age	1–15 yr	Dishwasher: 0.72–1.70 kWh/cyc Washer: 0.15–0.45 kWh/cyc Dryer: 1.15–3.50 kWh/cyc	[30, 31, 32]

run], which does indeed mean that they are correlated. However, this is a feature of the dataset and of household emissions, and it is actually possible for the two criterion to diverge. Since environmental impact comes from actual CO₂, which varies based on the type of power plant generating the power, grid mix composition, renewable penetration, and regional interconnection flows [35], and energy cost comes from the time of day, gas peakers, demand charges, transmission congestion pricing, and utility revenue requirements [36]. While Pennsylvania does not usually have such divergence, there are other areas that do, such as Texas (ERCOT), where high daytime wind generation can drive electricity prices near zero while gas-fired backup units sustain elevated marginal emissions during evening demand peaks [37, 38].

A sensitivity analysis was conducted to test the stability of the results. Fluctuations of ± 0.05 were applied to the weight vector by perturbing each of the four criterion weights individually and redistributing the difference equally across the remaining three, plus one equal-weight scenario ($w_j = 0.25$ for all j). The results can be seen in the below table, Table 5.

Table 5: Ten weight perturbation scenarios for the sensitivity analysis. Ene = Energy Cost, Env = Environmental, Com = Comfort, Pra = Practicality. Weights rounded to four decimal places. Architecture τ values are pending the corrected metrics run.

Scenario	w_{Ene}	w_{Env}	w_{Com}	w_{Pra}
Baseline	0.3000	0.3500	0.2000	0.1500
Ene +0.05	0.3500	0.3333	0.1833	0.1333
Ene -0.05	0.2500	0.3667	0.2167	0.1667
Env +0.05	0.2833	0.4000	0.1833	0.1333
Env -0.05	0.3167	0.3000	0.2167	0.1667
Com +0.05	0.2833	0.3333	0.2500	0.1333
Com -0.05	0.3167	0.3667	0.1500	0.1667
Pra +0.05	0.2833	0.3333	0.1833	0.2000
Pra -0.05	0.3167	0.3667	0.2167	0.1000
Equal	0.2500	0.2500	0.2500	0.2500

The Environmental weight of 0.35 reflects two convergent considerations. Residential emissions account for a sizable share of national CO₂ emissions [14], so a household decision-support tool that does not give priority to environmental impact fails. The Value-belief-norm theory, which says that show that environmental orientation is the most consistent normative driver across pro-environmental household behaviors [39, 40, 41, 42, 43, 44, 45, 46]. Finally, anyone using such a tool obviously has environmental motivations, and entropy-based normalization of the scenario distributions assigns higher information weight to the environmental criterion, independently supporting an above-average allocation [47]. The sensitivity analysis (Table 5) confirms that the architecture ordering is preserved under ± 0.05 perturbations of this weight, indicating that the 0.35 choice is not a threshold effect.

Cost is the strongest independent predictor of consumer adoption. This is clear for multiple reasons. First, cost savings as shown on energy labels for consumer products were more effective than actual energy use and carbon emissions.[48]. When energy labels strip away concrete cost-savings figures, the influence of all remaining information drops sharply [49, 50]. Second, households misestimate energy use, with Attari et al. reporting that participants underestimated energy use and savings by a factor of 2.8 on average [26]. Thus, high importance must be given to savings, even if the user is automating the process. It would also be beneficial to directly display actual cost savings. Weighting Energy Cost at 0.30 keeps it slightly below environmental impact while keeping its role as the criterion households actually act on. For HVAC decisions, thermostat setpoints directly determine electricity draw at the billing rate; households shown real-time cost feedback respond by reducing peak-period consumption, confirming that cost remains the operative signal even for automated decisions [51]. For shower decisions, water heating accounts for a significant share of residential energy costs and the per-shower increment is rarely visible to users, amplifying the same underestimation bias and making cost-feedback interventions similarly effective [52, 53].

Comfort is weighted at 0.20. Thermal comfort is the dominant driver of HVAC behavior even when it conflicts with energy savings [54, 55, 56], and low-income households in particular trade comfort against cost in documented ways

[57]. Anchoring comfort to ASHRAE Standard 55 [58, 59] keeps the criterion physically interpretable for HVAC and provides a 0–10 scale that generalizes (with domain-specific calibration described in Section 2.3) to appliance delay and shower duration. For appliance decisions, comfort is primarily a function of run-time delay: demand-response research finds that household willingness to accept a shift decays sharply beyond two to four hours, with dishwashers tolerated longest and dryers shortest [60, 61], and the penalty scales further with household size as dishes and laundry accumulate faster. For shower decisions, comfort peaks around the REU2016 average of 7.8 minutes [52, 53] and depends additionally on temperature adequacy, with heater setpoints below CDC Legionella thresholds carrying a comfort penalty [62].

Practicality is weighted at 0.15. A persistent finding in the heuristics literature is that any tool that adds friction relative to the existing routine sees adoption drop sharply, and heuristic decision-making remains the default for everyday energy choices [15, 16, 63, 64]. Penalizing alternatives that violate budget, system, or behavioral limits keeps the tool’s recommendations in check to make sure that households would actually adopt it long term. The lower weight (relative to comfort) reflects that practicality is a constraint on what is reasonable rather than a primary preference dimension. Where comfort captures what a household would prefer given no constraints, practicality captures what is feasible and sustainable to adopt. A 15-minute shower may score near-optimally on comfort, but if four occupants share a 40-gallon tank, available hot water runs out before the third person showers — making that alternative infeasible regardless of comfort preference.

Table 6: Reference ranges (5th–95th percentiles). Shower Environmental Impact is reported in gallons of water; HVAC and Appliance in lbs CO₂.

Criterion	HVAC	Appliance	Shower
Energy Cost (\$)	0.47–3.31	0.017–1.12	0.15–1.50
Environmental Impact	2.42–18.14	0.244–3.644	7.5–75.0
Comfort	0.0–10.0	0.0–10.0	0.0–10.0
Practicality	0.5–10.0	0.5–10.0	0.5–10.0

The reference ranges in Table 6 are anchored to the ground-truth calculators and reflect the 5th–95th percentile of the scenario cost and emissions (or water-volume) distributions rather than extreme theoretical limits, so that normalization preserves meaningful spread across typical residential alternatives. Using a combination of researched dataset and tested dataset bounds rather than the full theoretical physics is a deliberate modeling choice [47]: it concentrates score sensitivity in the range households actually encounter, at the cost of truncating scores for extreme outlier scenarios (documented as a paper limitation in the Discussion).

For HVAC energy cost, the percentile envelope is evaluated at a standardized 8-hour window with Pennsylvania’s residential electricity rate (\$0.19/kWh) [7, 65] and anchored to residential HVAC efficiency and degradation studies (efficient systems: [66]; degraded systems: [67]). The HVAC environmental bounds are derived from the same kWh envelope mapped through PJM marginal emissions factors (0.976 off-peak, 1.041 peak lbs CO₂/kWh) [35], maintaining a physically consistent link between cost and emissions.

For appliances, the minimum cost bound combines a 5th-percentile per-cycle energy use of 0.25 kWh for efficient HE washers (ENERGY STAR certified-products distribution) [68] with the lowest off-peak utility rate (\$0.067/kWh, West Penn off-peak) [36], giving \$0.017; the maximum bound expands ENERGY STAR’s upper kWh/cycle benchmark from 2.82 to 3.5 kWh/cycle to cover older non-certified resistance dryers paired with the highest peak rate (\$0.320/kWh, PECO peak) [36], giving \$1.12 [69, 70, 71, 72, 73]. The 3.5 kWh/cycle ceiling extends the ENERGY STAR maximum by 0.68 kWh/cycle to account for pre-certification electric resistance dryers still in service in older PA housing stock; measured consumption for such units reported in northeast field studies [72] and heat-pump dryer performance comparisons [71] supports a distribution tail above the certified ceiling. The environmental range applies the same kWh envelope to PJM marginal factors, giving 0.244–3.644 lbs CO₂.

For showers, energy cost bounds (\$0.15–\$1.50) come from a low-flow short-duration summer case (1.5 GPM, 5 min, inlet 65 °F) and a higher-flow long-duration winter case (3.5 GPM, 15 min, inlet 45 °F), with seasonal inlet temperatures from NREL hot-water modeling [74], duration distributions from REU2016 [52] and The Harris Poll [53], and GPM benchmarks from EPA WaterSense Appendix B [29]. Because the mixing-fraction physics targets a fixed delivery temperature of 105 °F, shower energy depends only on the rise from the mains inlet to that target and is independent of heater setpoint; for the stated 3.5 GPM, 15 min winter case the physics-derived maximum is

\$1.59, and the reference ceiling is set conservatively at \$1.50 consistent with the dataset-percentile framing. Shower environmental impact is defined as water volume (gallons) using WaterSense 5th–95th percentile flow rates and 5–15 minute durations, giving the 7.5–75 gallon envelope. The 95th-percentile GPM bound of 5.0 GPM is taken from EPA WaterSense conservation benchmarks [29] and extends above the dataset maximum of 4.0 GPM, using the published reference upper bound rather than the sample ceiling.

2.3. Ground Truth Calculators

Each ground-truth calculator consumes a scenario with three alternatives and returns four scores per alternative—Energy Cost (\$), Environmental Impact, Comfort, and Practicality (the latter three on 0–10)—plus the raw physical quantities before value-function transformation. Energy cost is always in dollars per 8-hour decision period; Environmental Impact is in lbs CO₂ for HVAC and Appliance decisions and in gallons of water for Shower decisions. Full equation-by-equation derivations are in Appendix A.

For HVAC and Appliance decisions, environmental impact uses PJM marginal emissions factors:

$$\text{CO}_2 = E_{\text{kWh}} \cdot \varepsilon(t), \quad \varepsilon(t) = \begin{cases} 1.041 \text{ lbs/kWh} & \text{peak (7am–11pm)} \\ 0.976 \text{ lbs/kWh} & \text{off-peak} \end{cases} \quad (2)$$

where E_{kWh} is the decision-period energy use and $\varepsilon(t)$ is the PJM marginal emissions factor at run time t [35]. Marginal rather than average factors are used because shifting a residential load displaces or adds the generator at the margin. All three calculators apply an identical four-tier budget penalty $P(u)$ multiplicatively to the energy-cost value-function output, where $u = C_{\text{monthly}}/B$ is monthly cost utilization relative to the household’s stated budget B :

$$P(u) = \begin{cases} 1.0 & u < 0.80 \\ 1 - 2.5(u - 0.80) & 0.80 \leq u < 1.0 \\ 0.5 e^{-3(u-1.0)} & 1.0 \leq u < 1.5 \\ 0 & u \geq 1.5 \end{cases} \quad (3)$$

where $P(u)$ is a unitless multiplicative penalty on the (post value-function) energy-cost score, u is monthly budget utilization, C_{monthly} is estimated monthly energy/water-related cost for the decision type, B is the household-stated monthly budget for utilities, and e is Euler’s number. The no-penalty zone ($u < 0.80$) reflects mental budget safety margins [75]; the linear decline approaching the limit follows Heath & Soll’s budget self-control model [76]; exponential loss aversion for overruns follows Prelec & Loewenstein [77] and Heutel [78]; and elimination beyond 150% reflects financial infeasibility [79]. Applying the penalty multiplicatively ensures budget-infeasible alternatives cannot dominate MAVT ranking on energy cost alone.

2.3.1. HVAC Calculator

Thermal load decomposes into four ASHRAE-style components. For cooling:

$$Q_{\text{cool}} = \underbrace{\frac{m \cdot A}{R} \Delta T}_{\text{conductive}} + \underbrace{400n + A + 800}_{\text{internal}} + \underbrace{0.15 A \times 20}_{\text{solar}} + \underbrace{1.08 \frac{A h_c \alpha}{60} \Delta T}_{\text{ventilation}} \quad (4)$$

where A is floor area (sq ft); m is a housing-type envelope multiplier (1.7 single-family, 1.5 townhouse/rowhouse, 1.2 apartment/condo) [80]; R is the wall insulation R-value; $\Delta T = T_{\text{out}} - T_{\text{in}}$; n is household size; $h_c = 8$ ft and $\alpha = 0.35$ ACH for ceiling height and infiltration rate [80]. The 800 BTU/hr baseline captures always-on standby loads per ACCA Manual J [80]. The heating load uses the same expression with $\Delta T = T_{\text{in}} - T_{\text{out}}$, no solar term, and internal gains subtracted rather than added.

SEER degrades with age via a front-loaded schedule (1.5× base rate for the first ten years, 0.5× thereafter, capped at 30%) [81, 82, 83]. Effective EER is estimated from the age-adjusted SEER via the AHRI 210/240 quadratic [84]:

$$\text{EER}_{\text{eff}} = -0.02 \text{SEER}_{\text{eff}}^2 + 1.12 \text{SEER}_{\text{eff}} \quad (5)$$

where SEER_{eff} is the age-adjusted seasonal energy efficiency ratio and EER_{eff} is the resulting effective energy efficiency ratio used for the decision period. Energy consumption over the 8-hour decision window is then:

$$E_{\text{kWh}} = \frac{Q_{\text{load}}}{\text{EER}_{\text{eff}} \times 1000} \cdot h \cdot m_{\text{occ}} \quad (6)$$

where Q_{load} is the thermal load (BTU/hr) for the decision context, h is the decision-window duration (8 hr), and m_{occ} is an occupancy multiplier. where $h = 8$ hr and m_{occ} scales for occupancy context: 1.0 for fully occupied, 0.75 for overnight, and $1 - 0.5(h_{\text{away}}/24)$ for daytime unoccupied periods [85], where h_{away} is hours-away per 24-hour day.

Comfort uses a tent function anchored to ASHRAE 55 PMV-optimal setpoints [86, 87, 88]:

$$C_{\text{comfort}} = \max(0, 10 - |T_{\text{in}} - T^*|) \quad (7)$$

where C_{comfort} is a 0–10 comfort score, T_{in} is the indoor setpoint, and T^* is the context-dependent optimal setpoint used as the peak of the tent function. where $T^* = 76$ °F in cooling mode ($T_{\text{out}} > 75$ °F) and 70 °F in heating mode, consistent with ASHRAE 55-2020 sedentary PMV bands [86]. A small penalty scales with household size beyond three occupants. Practicality penalizes extreme setpoints and large outdoor–indoor ΔT (signaling operation near system limits) via multipliers calibrated to equipment-stress thresholds.

2.3.2. Appliance Calculator

Energy cost per cycle is $C = E_{\text{cyc}} \cdot r(t, \ell)$, where E_{cyc} is kWh/cycle and $r(t, \ell)$ is the location- and time-specific TOU rate resolved to one of six Pennsylvania utilities (PECO, PPL, West Penn, Penelec, MetEd, Duquesne) [36]. where C is per-cycle energy cost (dollars/cycle), E_{cyc} is per-cycle energy use (kWh/cycle), t is the appliance run time (time-of-day), and ℓ is the household location used to resolve the serving utility. The PJM emissions window (7am–11pm) applies to environmental impact regardless of the utility’s billing window, because emissions follow the regional grid. The run-time rate period is selected by mapping location ℓ to a utility $U(\ell)$ and then applying that utility’s TOU peak window otherwise.

$$(h_{\text{pk},s}^U, h_{\text{pk},e}^U) : r(t, \ell) = r_{\text{pk}}^U \text{ if } h(t) \in [h_{\text{pk},s}^U, h_{\text{pk},e}^U) \text{ and } r(t, \ell) = r_{\text{off}}^U \quad (8)$$

where $U(\ell)$ maps location ℓ to a serving utility, $h_{\text{pk},s}^U$ and $h_{\text{pk},e}^U$ are that utility’s TOU peak-window start and end hours, r_{pk}^U and r_{off}^U are the utility’s peak and off-peak energy prices (\$/kWh), and $h(t)$ maps time t to an hour-of-day on a 24-hour clock. In the implementation, the six utilities use distinct peak windows (e.g., 2–6pm for PECO and PPL, 2–9pm for most FirstEnergy utilities). Comfort is a piecewise decay from 10 at zero delay, with appliance-specific tolerance ceilings of 12 hr (dishwashers), 8 hr (washers), and 6 hr (dryers) [60, 61], plus an additive noise penalty scaled by housing type for runs after 10 pm [89] and a household-size penalty. Delays are computed as the minimum circular distance from the baseline time t_0 on a 24-hour clock (avoiding wrap-around artifacts), i.e.,

$$d(t, t_0) = \min((h(t) - h(t_0)) \bmod 24, (h(t_0) - h(t)) \bmod 24), \quad (9)$$

where $d(t, t_0)$ is delay in hours between run time t and baseline time t_0 , $h(\cdot)$ maps a time to hour-of-day, and mod24 indicates wrap-around on a 24-hour clock. The base comfort component uses the following piecewise schedule on delay hours d :

$$C_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - \frac{2}{3}d & 0 < d \leq 3 \\ 8 - \frac{1}{2}(d - 3) & 3 < d \leq 7 \\ 6 - \frac{2}{3}(d - 7) & 7 < d \leq 12 \\ 2 & d > 12 \end{cases} \quad (10)$$

where $C_{\text{base}}(d)$ is the pre-penalty comfort score (0–10) as a function of delay d . The late-night noise penalty is applied when running between 10pm and 7am and the appliance exceeds a dBA threshold:

$$N(t, \text{type}, \text{housing}) = \begin{cases} 0 & \text{if } h(t) \in [7, 22) \text{ or } \text{dBA}(\text{type}) \leq \theta \\ 2 \cdot \kappa(\text{housing}) & \text{otherwise} \end{cases} \quad (11)$$

where $N(t, \text{type}, \text{housing})$ is a nonnegative noise-disruption penalty in comfort-score points, $\text{dBA}(\text{type})$ is the appliance noise level for the appliance type, θ is the dBA threshold, and $\kappa(\text{housing})$ is a unitless housing-type multiplier (larger for shared-wall housing). where $\theta = 45$ dBA and $\kappa(\cdot)$ is a housing-type multiplier (larger for shared-wall housing). The household-size penalty scales with delay relative to the appliance-specific maximum acceptable delay $d_{\max}(\text{type})$:

$$S(d, n, \text{type}) = \sigma(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \quad (12)$$

where $S(d, n, \text{type})$ is a nonnegative household-size penalty in comfort-score points, n is occupants, $d_{\max}(\text{type})$ is the maximum acceptable delay (hours) for the appliance type, and $\sigma(n)$ is an occupants-based factor (higher for larger households). where $\sigma(n)$ is an occupants-based factor (higher for larger households). Final comfort is then $C_{\text{comfort}} = \text{clip}_{[0,10]}(C_{\text{base}} - N - S)$. where $\text{clip}_{[a,b]}(x) = \min(b, \max(a, x))$ clamps a value to the interval $[a, b]$. Practicality mirrors the delay schedule with additional penalties for late-night timing complexity and household coordination difficulty. The base practicality component is:

$$P_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - d & 0 < d \leq 2 \\ 8 - \frac{3}{4}(d - 2) & 2 < d \leq 4 \\ 6.5 - \frac{1}{2}(d - 4) & 4 < d \leq 8 \\ 4.5 - \frac{3}{8}(d - 8) & 8 < d \leq 12 \\ 1.5 & d > 12 \end{cases} \quad (13)$$

where $P_{\text{base}}(d)$ is the pre-penalty practicality score (0–10) as a function of delay d . Timing complexity applies an additive penalty by run-time hour:

$$T(t) = \begin{cases} 2 & 0 \leq h(t) < 6 \\ 1 & 22 \leq h(t) < 24 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $T(t)$ is a nonnegative timing-complexity penalty in practicality-score points and $h(t)$ is hour-of-day. Household coordination difficulty uses the same delay scaling form as in comfort:

$$K(d, n, \text{type}) = \kappa_n(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \quad (15)$$

where $K(d, n, \text{type})$ is a nonnegative household-coordination penalty in practicality-score points and $\kappa_n(n)$ is an occupants-based factor increasing with occupants. with $\kappa_n(n)$ increasing with occupants. Final practicality is $C_{\text{prac}} = \text{clip}_{[1.5,10]}(P_{\text{base}} - T - K)$, with a daytime floor applied for $h(t) \in [7, 22)$ and $d \leq d_{\max}(\text{type})$. where C_{prac} is the final 0–10 practicality score (after penalties and clipping). For budget utilization in (3), the appliance calculator converts per-cycle cost to a monthly estimate as:

$$C_{\text{monthly}} = 30 C_{\text{cycle}}. \quad (16)$$

where C_{cycle} is per-cycle energy cost and the factor 30 assumes 30 cycles per month.

2.3.3. Shower Calculator:

Mains inlet temperature is interpolated from outdoor temperature between 45 °F (winter, $T_{\text{out}} \leq 32$ °F) and 65 °F (summer, $T_{\text{out}} \geq 75$ °F) [74, 90]. Only the fraction of flow requiring heating reaches the heater; that mixing fraction is:

$$f_{\text{hot}} = \frac{T_{\text{target}} - T_{\text{inlet}}}{T_{\text{heater}} - T_{\text{inlet}}} \quad (17)$$

where f_{hot} is the fraction of total flow drawn as hot water, T_{target} is desired delivery temperature, T_{inlet} is mains inlet temperature, and T_{heater} is the water-heater setpoint temperature. where $T_{\text{target}} = 105^\circ\text{F}$ is the standard comfortable delivery temperature [91]. Shower energy is then:

$$E_{\text{kWh}} = \frac{\dot{V} \cdot f_{\text{hot}} \cdot 8.33 \cdot (T_{\text{heater}} - T_{\text{inlet}}) \cdot \tau}{3412 \cdot \eta} \quad (18)$$

where E_{kWh} is shower energy use (kWh), $\dot{V}$ is total flow rate (GPM), τ is shower duration (min), η is electric heater efficiency (UEF), and 3412 is the BTU-per-kWh conversion; the constant 8.33 converts gallons to pounds of water (lb/gal). where $\dot{V}$ is flow rate (GPM), 8.33 lb/gal is water density, τ is shower duration (min), 3412 BTU/kWh is the standard conversion, and $\eta = 0.92$ is the unified energy factor for a 40–55 gal electric tank [92]. Environmental impact is water volume: $V = \dot{V} \cdot \tau$ (gallons). where V is water volume used (gallons). Energy cost uses a flat electricity rate p_e :

$$C_{\$} = p_e E_{\text{kWh}}. \quad (19)$$

where $C_{\$}$ is shower energy cost (dollars) and p_e is the electricity price (\$/kWh). Comfort is a piecewise function peaking at the REU2016 average of 7.8 min [52, 53], with additive penalties for temperature adequacy (CDC Legionella thresholds [62]) and household contention. The implementation uses a duration-based piecewise base comfort $C_{\text{base}}(\tau)$ (with a short-duration cubic ramp to reflect severely rushed showers), then subtracts a temperature penalty $\Delta_{\text{temp}}(T_{\text{heater}})$ and a household contention penalty $\Delta_{\text{cont}}(\tau, n)$, and clips to $[0, 10]$:

$$C_{\text{comfort}}(\tau, T_{\text{heater}}, n) = \text{clip}_{[0,10]}(C_{\text{base}}(\tau) - \Delta_{\text{temp}}(T_{\text{heater}}) - \Delta_{\text{cont}}(\tau, n)). \quad (20)$$

where τ is shower duration (min), n is occupants, $C_{\text{base}}(\tau)$ is the base comfort curve, $\Delta_{\text{temp}}(T_{\text{heater}})$ is a temperature-adequacy penalty, and $\Delta_{\text{cont}}(\tau, n)$ is a household-contention penalty. Practicality additionally penalizes alternatives that exceed available tank capacity. The practicality model uses a duration-based base practicality $P_{\text{base}}(\tau)$ and a capacity constraint computed from hot-water demand. Hot-water per shower is $\tau \dot{V} f_{\text{hot}}$ (gal), total household hot-water need is $n \tau \dot{V} f_{\text{hot}}$, and available draw is approximated as 0.8 of nominal tank size S :

$$H_{\text{need}} = n \tau \dot{V} f_{\text{hot}}, \quad H_{\text{avail}} = 0.8 S. \quad (21)$$

where H_{need} is total hot-water volume required for the household’s showers over the relevant period (gal), H_{avail} is available hot-water draw volume (gal), and S is nominal tank capacity (gal). A fixed penalty is applied when $H_{\text{need}} > H_{\text{avail}}$, yielding a clipped final practicality score. For budget utilization in (3), shower monthly cost is estimated from per-shower cost using an average showers-per-person-per-day rate:

$$C_{\text{monthly}} = C_{\text{shower}} \cdot (n \cdot 0.9 \cdot 30). \quad (22)$$

where C_{shower} is per-shower energy cost, n is occupants, 0.9 is showers/person/day, and 30 is days/month.

2.4. Architecture Implementations

All three architectures call the same underlying model via OpenRouter: Mistral Small 3.2 24B Instruct at temperature 0.3. Temperature is fixed so the comparison isolates architecture from sampling variance. Each architecture is described below; pipeline schematics appear in Figure 4.

2.4.1. Pure Prompting

The Pure Prompting architecture issues one API call per alternative (three calls per scenario) using a calibrated system prompt that combines role-prompting [93], structured-output constraints [94], and explicit bias-mitigation language drawn from Galaz et al. on AI sustainability priors [95]. The model is asked to return JSON with all four criterion scores on a 0–10 scale; no external context is supplied beyond the scenario description.

2.4.2. RAG-Enhanced

The RAG-Enhanced architecture retrieves the top $k = 3$ most similar scenarios from a ChromaDB vector store before scoring, using the sentence-transformers/all-MiniLM-L6-v2 embedding model. The retrieved scenarios and their ground-truth scores are injected into the system prompt as context; the model is instructed to use them as reference but to score the target scenario independently. Like Pure Prompting, RAG-Enhanced uses three API calls per scenario (one per alternative). The retrieval index is built once from the 231 RAG-only scenarios described in Section 2.1 and is never queried with test scenarios [25].

2.4.3. Hybrid

The Hybrid architecture issues a single API call per scenario to extract structured parameters (decision type, calculator inputs such as SEER tier, kWh/cycle, GPM, baseline time, occupancy context, and the indoor temperatures or run times or durations that define the alternatives). The extracted JSON is validated, passed to the corresponding ground-truth calculator, and the four MAVT scores are returned. Hybrid uses 1 API call per scenario versus 3 for Pure and RAG and exploits the principle of decomposed prompting [96], separating the natural-language understanding step (the LLM’s strength) from the quantitative scoring step (the calculator’s strength).

2.5. Evaluation Metrics

Accuracy is evaluated on two axes. Score-level error uses Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) between architecture scores and ground-truth scores on the 0–10 MAVT scale, computed separately for each criterion and aggregated across criteria. Ranking accuracy uses Kendall’s τ and Spearman’s ρ between the architecture’s induced ranking of the three alternatives and the ground-truth ranking, together with Top-1 accuracy (architecture’s best-ranked alternative matches ground truth) and Top-2 accuracy (ground-truth top alternative appears in the architecture’s top two). Efficiency is evaluated using total tokens (input plus output) and total latency, both summed across the 181-scenario test set. Failure rate is reported as the fraction of scenarios with malformed output, missing scores, out-of-bounds scores, or invalid score types. Scenario matching between architectures and ground truth uses (question, location) content keys to handle the case where scenario IDs drift across runs.

2.6. Architectures

Figure 3 reproduces the Dietz et al. display of behavioral plasticity, net carbon effect, and the Reasonably Achievable Emission Reduction (RAER) per action.

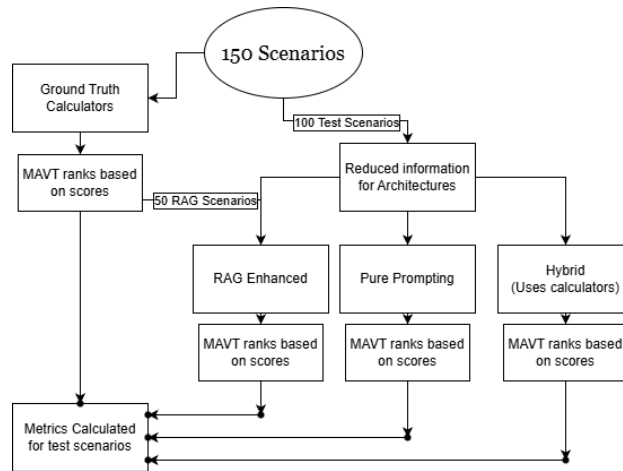


Figure 3:

3. Results

[BLOCKED in this draft: see audit log at end of file. Results 3.1–3.4 (overall ranking accuracy, criterion-level error, performance by decision type, token usage and latency) are not written here because the underlying metrics-summary numbers shown in Items 5–13 are flagged as incorrect by the project lead and a corrected multi-model run is pending.]

4. Discussion

4.1. Limitations

The results reported in this study derive exclusively from a single model, Mistral Small 3.2 24B, via OpenRouter. Although the multi-model benchmark pipeline is fully implemented for four models—Qwen 3.5 9B, Qwen 3.5 27B, Gemini 3.5 Flash, and GPT-5—[CONFIRM AFTER RUNNING: update whether architecture-level conclusions can be separated from model-level effects and whether the Hybrid advantage holds across model families].

A multi-run protocol of $N = 10$ runs per model is defined in the codebase and standard deviation columns are implemented in `CalculateMetrics.py`. [CONFIRM AFTER RUNNING: report observed variance across runs and assess whether performance gaps between architectures are statistically significant.] Until then, all reported metrics—Kendall’s τ , Top-1 accuracy, MAE—reflect a single run at temperature $T = 0.3$ and are not accompanied by confidence intervals.

The most critical validity threat is that the same researcher designed both the ground truth calculators and the Hybrid architecture. Hybrid’s advantage may partly reflect design alignment—the calculator the Hybrid calls is the same one that produces the ground truth—rather than a generalizable architectural superiority. This threat cannot be eliminated within the current study design and must be weighed when interpreting the magnitude of Hybrid’s performance gap over RAG-Enhanced and Pure Prompting.

Results are specific to Pennsylvania’s grid and utility rate structure. The PJM marginal emissions factors (peak: 1.041 lbs CO₂/kWh; off-peak: 0.976 lbs CO₂/kWh) and Pennsylvania time-of-use rate tiers used in the ground truth calculators do not generalize to high-renewables grids, where lower and more variable emissions intensities would compress environmental score distributions and potentially change which architecture performs best.

The environmental criterion for shower decisions uses water consumption in gallons as a proxy rather than CO₂ emissions. This makes the environmental dimension semantically heterogeneous across decision types: HVAC and Appliance environmental scores reflect grid emissions, while Shower environmental scores reflect water volume. Cross-decision-type comparisons of environmental MAE should be interpreted with this distinction in mind; the proxy does not affect within-type ranking validity.

Prompt consistency across Pure Prompting and RAG-Enhanced has not been fully verified. A shared prompt utility module is not present in the repository, meaning the two architectures use independently maintained prompt strings. Divergence between those strings cannot be excluded without a unified implementation, introducing an uncontrolled variable in the comparison.

4.2. Future Work

The immediate priority is completing the multi-model benchmark. The pipeline is fully configured for Qwen 3.5 9B, Qwen 3.5 27B, Gemini 3.5 Flash, and GPT-5, representing a 2×2 grid of open versus closed-weight and small versus large model classes. [CONFIRM AFTER RUNNING: state whether the Hybrid advantage is robust or model-dependent, and summarize architecture-level conclusions across all four models.]

The per-run file naming infrastructure and standard deviation columns in `CalculateMetrics.py` are already implemented for $N = 10$ runs per model. [CONFIRM AFTER RUNNING: report whether Hybrid’s deterministic calculator component measurably suppresses run-to-run variance relative to the two fully LLM-driven architectures.]

The Hybrid architecture shows a pronounced performance drop from HVAC ($\tau = 0.90$) to Appliance ($\tau = 0.68$) and Shower ($\tau = 0.60$). Scenario-type-specific extraction prompts for Appliance and Shower decisions should be developed to address this gap, as the current extraction prompt was optimized primarily against HVAC scenarios.

Replacing the water-consumption proxy in the Shower environmental criterion with a lifecycle CO₂ factor—converting gallons of hot water to grid emissions using a standardized heating efficiency—would make the environmental criterion semantically consistent across all three decision types and enable valid cross-type environmental MAE comparisons.

5. Conclusion

[BLOCKED in this draft: see audit log. The Conclusion is gated on corrected results.]

References

- [1] U. E. P. Agency, Assumptions and references for household carbon footprint calculator, Tech. rep., U.S. Environmental Protection Agency, Washington, DC. (2024).
- [2] W. contributors, Domestic energy consumption, section: Average domestic energy consumption per household in the United States (2024).
- [3] C. M. Jones, D. M. Kammen, Quantifying carbon footprint reduction opportunities for U.S. households and communities, *Environmental Science & Technology* 45 (9) (2011) 4088–4095. doi:10.1021/es102221h.
- [4] U. o. M. Center for Sustainable Systems, Carbon footprint factsheet, Tech. rep., Center for Sustainable Systems, University of Michigan, pub. No. CSS09-05 (2023).
- [5] A. M. Ohler, D. G. Loomis, K. Ilves, A study of electricity savings from energy star appliances using household survey data, *Energy Policy* 144 (2020) 111607. doi:10.1016/j.enpol.2020.111607.
- [6] U. D. of Energy, Why energy efficiency matters, U.S. Department of Energy webpage.
- [7] U. E. I. Administration, Frequently asked questions (faqs), Tech. rep., U.S. Energy Information Administration, 2022. (2022).
- [8] B. Goldstein, D. Gounaridis, J. P. Newell, The carbon footprint of household energy use in the United States, *Proceedings of the National Academy of Sciences* 117 (32) (2020) 19122–19130. doi:10.1073/pnas.1922205117.
- [9] A. Kollmuss, J. Agyeman, Mind the gap: Why do people act environmentally and what are the barriers to pro-environmental behavior?, *Environ. Educ. Res.* 8 (3) (2002) 239–260. doi:10.1080/13504620220145401.
- [10] W. Collinge, A. Landis, A. Jones, L. Schaefer, M. Bilec, Productivity metrics in dynamic LCA for whole buildings: Using a post-occupancy evaluation of energy and indoor environmental quality tradeoffs, *Build. Environ.* 82 (2014) 339–348. doi:10.1016/j.buildenv.2014.08.032.
- [11] I. Huang, J. Keisler, I. Linkov, Multi-criteria decision analysis in environmental sciences: Ten years of applications and trends, *Sci. Total Environ.* 409 (19) (2011) 3578–3594. doi:10.1016/j.scitotenv.2011.06.022.
- [12] J. Cernevičienė, A. Kabasinskias, Review of multi-criteria decision-making methods in finance using explainable artificial intelligence, *Front. Artif. Intell.* 5 (2022). doi:10.3389/frai.2022.827584.
- [13] H. Huang, P. Burgherr, Mcda calculator: A streamlined decision support system for multi-criteria decision analysis, Tech. rep., H. Huang and P. Burgherr, in: S.P. Duarte, et al. (Eds.), *Decision Support Systems XIV. ICDSST 2024. Lecture Notes in Business Information Processing*, vol. 506, Springer, Cham, 2024. (2024).
- [14] T. Dietz, G. Gardner, J. Gilligan, P. Stern, M. Vandenbergh, Household actions can provide a behavioral wedge to rapidly reduce US carbon emissions, *Proc. Natl. Acad. Sci. U.S.A.* 106 (44) (2009) 18452–18456. doi:10.1073/pnas.0908738106.
- [15] D. Ariely, D. Zakay, A timely account of the role of duration in decision making, *Acta Psychol.* 108 (2) (2001) 187–207. doi:10.1016/S0001-6918(01)00034-8.
- [16] F. Leclerc, B. Schmitt, L. Dube, Waiting time and decision making: Is time like money?, *J. Consum. Res.* 22 (1) (1995) 110–119. doi:10.1086/209439.
- [17] F.-H. Nah, A study on tolerable waiting time: How long are web users willing to wait?, *Behav. Inf. Technol.* 23 (3) (2004) 153–163. doi:10.1080/01449290410001669914.
- [18] S. Bobadilla-Suarez, B. C. Love, Fast or frugal, but not both: Decision heuristics under time pressure, *Journal of Experimental Psychology: Learning, Memory, and Cognition* 44 (1) (2018) 24–33. doi:10.1037/xlm0000419.

- [19] M. Hahn, R. Lawson, Y. Lee, The effects of time pressure and information load on decision quality, *Psychol. Mark.* 9 (5) (1992). doi:10.1002/mar.4220090503.
- [20] G. Phillips-Wren, M. Adya, Decision making under stress: The role of information overload, time pressure, complexity, and uncertainty, *J. Decis. Syst.* 29 (sup1) (2020) 213–225. doi:10.1080/12460125.2020.1768680.
- [21] I. Jackson, D. Ivanov, A. Dolgui, J. Namdar, Generative artificial intelligence in supply chain and operations management: A capability-based framework for analysis and implementation, *Int. J. Prod. Res.* 62 (2024) 6120–6145. doi:10.1080/00207543.2024.2309309.
- [22] M. Huang, R. Rust, A strategic framework for artificial intelligence in marketing, *J. Acad. Mark. Sci.* 49 (2020) 30–50. doi:10.1007/s11747-020-00749-9.
- [23] M. Simon, S. Kamin, A. Hamper, T. Wittenberg, S. Schmitt-Ruth, Strategizing ai in healthcare: A multidimensional blueprint for transformative decision-making in clinical settings, *Tech. rep.*, M. Simon and S. Kamin and A. Hamper and T. Wittenberg and S. Schmitt-Ruth, *curr. Dir. Biomed. Eng.* (2024). (2024). doi:10.1515/cdbme-2024-2146.
- [24] I. Svoboda, D. Lande, Enhancing multi-criteria decision analysis with ai: Integrating analytic hierarchy process and gpt-4 for automated decision support, *arXiv preprint arXiv:2402.07404* (2024). doi:10.48550/arXiv.2402.07404.
- [25] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive nlp tasks, in: *Advances in Neural Information Processing Systems (NeurIPS)* 33, 2020.
- [26] S. Attari, M. DeKay, C. Davidson, W. B. de Bruin, Public perceptions of energy consumption and savings, *Proc. Natl. Acad. Sci. U.S.A.* 107 (37) (2010) 16054–16059. doi:10.1073/pnas.1001509107.
- [27] C. E. Commission, Ja4.3 walls, *Tech. rep.*, California Energy Commission, 2013 Residential Compliance Manual: Joint Appendix 4 (2013).
- [28] U.S. Environmental Protection Agency, U.S. Department of Energy, Recommended home insulation r-values, *Tech. rep.*, ENERGY STAR, based on 2021 IECC Residential Provisions (2023).
- [29] U. E. P. Agency, Watersense specification for showerheads, version 1.1, *Tech. rep.*, U.S. Environmental Protection Agency, ePA WaterSense, July 26, 2018. (2018).
- [30] U.S. Environmental Protection Agency, Energy star dishwashers: Key product criteria (2023).
- [31] U.S. Environmental Protection Agency, Energy star certified residential clothes washers dataset (2023).
- [32] U.S. Environmental Protection Agency, Energy star scoping report: Residential clothes dryers, *Tech. rep.*, ENERGY STAR (2023).
- [33] J. Dyer, R. Sarin, Measurable multiattribute value functions, *Oper. Res.* 27 (4) (1979) 810–822. doi:10.1287/opre.27.4.810.
- [34] R. Keeney, H. Raiffa, *Decisions with Multiple Objectives: Preferences and Value Trade-offs*, Wiley, New York, 1976.
- [35] PJM Interconnection, 2022 co2 emissions and generation resource mix for pjm and miso, accessed 2025-05-28 (2023).
- [36] P. E. Company, How time-of-use pricing works with interconnection to solar or net metering, *Tech. rep.*, PECO Energy Company, tOU customer documentation, 2021. (2021).

- [37] E. R. C. of Texas, Impact of increased wind resources in the ERCOT region, Tech. rep., ERCOT Market Operations and Market Design Departments (2019).
- [38] S. P. Holland, E. T. Mansur, N. Z. Muller, A. J. Yates, Spatial and temporal heterogeneity of marginal emissions: Implications for electric cars and other electricity-shifting policies, Tech. rep., National Bureau of Economic Research, nBER Working Paper No. 18462 (2016).
- [39] P. Stern, T. Dietz, T. Abel, G. Guagnano, L. Kalof, A value-belief-norm theory of support for social movements: The case of environmentalism, *Hum. Ecol. Rev.* 6 (2) (1999) 81–97.
- [40] F. Fornara, P. Pattitoni, M. Mura, E. Strazzera, Predicting intention to improve household energy efficiency: The role of value-belief-norm theory, normative and informational influence, and specific attitude, *J. Environ. Psychol.* 45 (2016) 1–10. doi:10.1016/j.jenvp.2015.11.001.
- [41] Y. Hong, A. A. Mamun, M. Masukujjaman, Q. Yang, Significance of the environmental value-belief-norm model and its relationship to green consumption among Chinese youth, *Asia Pac. Manag. Rev.* 29 (1) (2024) 127–140. doi:10.1016/j.apmr.2023.10.002.
- [42] L. Steg, L. Dreijerink, W. Abrahamse, Factors influencing the acceptability of energy policies: A test of vbn theory, *J. Environ. Psychol.* 25 (4) (2005) 415–425. doi:10.1016/j.jenvp.2005.08.003.
- [43] S. Lee, Y. Kim, T. Roh, Pro-environmental behavior on electric vehicle use intention: Integrating value-belief-norm theory and theory of planned behavior, *J. Clean. Prod.* 418 (2023) 138211. doi:10.1016/j.jclepro.2023.138211.
- [44] K. Rezaei-Moghaddam, N. Vatankhah, A. Ajili, Adoption of pro-environmental behaviors among farmers: Application of value-belief-norm theory, *Chem. Biol. Technol. Agric.* 7 (1) (2020). doi:10.1186/s40538-019-0174-z.
- [45] C.-P. Wang, Q. Zhang, P.P.W. Wong, L. Wang, Consumers' green purchase intention to visit green hotels: A value-belief-norm theory perspective, *Front. Psychol.* 14 (2023). doi:10.3389/fpsyg.2023.1139116.
- [46] H. Lind, T. Nordfjaern, S. Jorgensen, T. Rundmo, The value-belief-norm theory, personal norms and sustainable travel mode choice in urban areas, *J. Environ. Psychol.* 44 (2015) 119–125. doi:10.1016/j.jenvp.2015.06.001.
- [47] E. Roszkowska, Entropy and normalization in mcda: A data-driven perspective on ranking stability, *Entropy* 28 (1) (2026) 114, PMID: 41594021; PMCID: PMC12839561. doi:10.3390/e28010114.
- [48] R. Newell, J. Siikamaki, Nudging energy efficiency behavior: The role of information labels, *J. Assoc. Environ. Resour. Econ.* 1 (4) (2014) 555–598. doi:10.1086/679281.
- [49] H. Allcott, D. Taubinsky, Evaluating behaviorally motivated policy: Experimental evidence from the lightbulb market, *Am. Econ. Rev.* 105 (8) (2015) 2501–2538. doi:10.1257/aer.20131564.
- [50] M. Andor, A. Gerster, S. Sommer, Consumer inattention, heuristic thinking and the role of energy labels, *Energy J.* 41 (1) (2020) 83–112. doi:10.5547/01956574.41.1.mand.
- [51] Field experimental evidence on the effect of pricing on residential electricity consumption, *management Science*, online ahead of print (2023). doi:10.1287/mnsc.2020.02074.
- [52] W. DeOreo, P. Mayer, B. Dziegielewski, J. Kiefer, Residential end uses of water, version 2: Executive report, Tech. rep., Water Research Foundation, project 4309. (2016).
- [53] T. H. Poll, Examining Americans' shower habits, Tech. rep., The Harris Poll, April 30, 2024. (2024).
- [54] N. Aqilah, H. Rijal, S. Zaki, A review of thermal comfort in residential buildings: Comfort trends and energy saving potential, *Energies* 15 (23) (2022) 9012. doi:10.3390/en15239012.

- [55] S. Karjalainen, Gender differences in thermal comfort and use of thermostats in everyday thermal environments, *Build. Environ.* 42 (4) (2007) 1594–1603. doi:10.1016/j.buildenv.2006.01.009.
- [56] Z. Wang, T. Hong, Learning occupants' indoor comfort temperature through a bayesian inference approach for office buildings in united states, *Renew. Sustain. Energy Rev.* 119 (2020) 109593. doi:10.1016/j.rser.2019.109593.
- [57] C. Chen, X. Xu, J. Day, Thermal comfort or money saving? exploring intentions to conserve energy among low-income households in the united states, *Energy Res. Soc. Sci.* 26 (2017) 61–71. doi:10.1016/j.erss.2017.01.009.
- [58] ASHRAE, Standard 55 – thermal environmental conditions for human occupancy, aSHRAE Standard 55.
- [59] R. de Dear, G. Brager, Thermal comfort in naturally ventilated buildings: Revisions to ashrae standard 55, *Energy Build.* 34 (2002) 549–561. doi:10.1016/S0378-7788(02)00005-1.
- [60] A. Paetz, E. Dutschke, W. Fichtner, Shifting dish-washer use: A field study on time-of-use tariffs and user acceptance, in: ACEEE Summer Study on Energy Efficiency in Buildings, Paper 0193-000232, 2012, 2012.
- [61] M. Waseem, Z. Lin, S. Liu, I. Sajjad, T. Aziz, Optimal gwcso-based home appliances scheduling for demand response considering end-users comfort, *Electr. Power Syst. Res.* 187 (2020) 106477. doi:10.1016/j.epsr.2020.106477.
- [62] C. for Disease Control, Prevention, Monitoring building water: A vital step for control of legionella, Tech. rep., Centers for Disease Control and Prevention, 2026. (2026).
- [63] B. Gardner, P. Lally, A. Rebar, Does habit weaken the relationship between intention and behaviour? revisiting the habit-intention interaction hypothesis, *Soc. Pers. Psychol. Compass* (2020). doi:10.1111/spc3.12553.
- [64] E. Mergelsberg, B. Mullan, V. Allom, A. Scott, An intervention designed to investigate habit formation in a novel health behaviour, *Psychol. Health* 36 (2020) 405–426. doi:10.1080/08870446.2020.1779272.
- [65] U. E. I. Administration, Average price of electricity to ultimate customers by end-use sector, by state, 2024 and 2023 (table 2.10), Tech. rep., Electric Power Annual 2024 (2025).
- [66] S. E. Economics, A study of heat pump performance in u.s. cities, Tech. rep., Synapse Energy Economics (2013).
- [67] D. S. Parker, J. R. Sherwin, T. Stedman, M. Anello, Evaluation of air conditioning performance degradation, Tech. rep., Florida Solar Energy Center, fSEC-CR-2080-18 (2018).
- [68] U.S. Environmental Protection Agency, Energy star products, accessed 2025-05-28 (2024).
- [69] J. Chen-Yu, J. Emmel, Comparisons of fabric care performances between conventional and high-efficiency washers and dryers, *Fash. Text.* 5 (2018) 1–19. doi:10.1186/s40691-018-0135-3.
- [70] G. Hustvedt, M. Ahn, J. Emmel, The adoption of sustainable laundry technologies by us consumers, *Int. J. Consum. Stud.* 37 (2013) 291–298. doi:10.1111/ijcs.12007.
- [71] D. Winfield, et al., Measured performance of heat pump clothes dryers, in: ACEEE Summer Study on Energy Efficiency in Buildings, 2016, 2016.
- [72] N. E. E. Partnerships, New study unearths energy baseline for clothes dryers in the northeast, 2015 (2015).
- [73] G. Porras, G. Keoleian, G. Lewis, N. Seeba, A guide to household manual and machine dishwashing through a life cycle perspective, *Environ. Res. Commun.* 2 (2020). doi:10.1088/2515-7620/ab716b.
- [74] R. Hendron, J. Burch, Development of standardized domestic hot water event schedules for residential buildings, Tech. rep., R. Hendron and J. Burch, nREL/TP-550-40874, National Renewable Energy Laboratory, 2008. (2008). doi:10.1115/ES2007-36104.

- [75] R. Thaler, Mental accounting matters, *J. Behav. Decis. Making* 12 (3) (1999) 183–206. doi:10.1002/(SICI)1099-0771(199909)12:3<183::AID-BDM318>3.0.CO;2-F.
- [76] C. Heath, J. Soll, Mental budgeting and consumer decisions, *J. Consum. Res.* 23 (1) (1996) 40–52. doi:10.1086/209465.
- [77] D. Prelec, G. Loewenstein, The red and the black: Mental accounting of savings and debt, *Mark. Sci.* 17 (1) (1998) 4–28. doi:10.1287/mksc.17.1.4.
- [78] G. Heutel, Prospect theory and energy efficiency, Tech. rep., G. Heutel, nBER Working Paper No. 23692, National Bureau of Economic Research, 2017. (2017). doi:10.3386/w23692.
- [79] J. Gathergood, Self-control, financial literacy and consumer over-indebtedness, *J. Econ. Psychol.* 33 (3) (2012) 590–602. doi:10.1016/j.joep.2011.11.006.
- [80] B. Polly, M. Gestwick, M. Bianchi, Strategy guideline: Accurate heating and cooling load calculations, Tech. rep., National Renewable Energy Laboratory, nREL/TP-5500-51603 (2011).
- [81] P. Domanski, H. Henderson, W. Payne, Sensitivity analysis of installation faults on heat pump performance, Tech. rep., P.A. Domanski and H.I. Henderson and W.V. Payne, nIST Technical Note 1848, National Institute of Standards and Technology, 2014. (2014).
- [82] D. E. Group, Hvac energy efficiency maintenance study, Tech. rep., Davis Energy Group, prepared for California Investor-Owned Utilities (CALMAC ID HVAC_EE_MAINTENANCE_FINAL), 2010. (2010).
- [83] K. Fenaughty, D. Parker, Evaluation of air conditioning performance degradation: Opportunities from diagnostic methods, FSEC-PF-474-18, Florida Solar Energy Center, 2018 (2018).
- [84] Air-Conditioning, Heating, and Refrigeration Institute (AHRI), Ansi/ahri standard 210/240-2008: Performance rating of unitary air-conditioning and air-source heat pump equipment (2008).
- [85] S. Houde, E. Myers, A. Siddiqui, Smart thermostats, automation, and time-varying prices, Tech. rep., Resources for the Future, rFF Working Paper 21-20 (2021).
- [86] ASHRAE, Ansi/ashrae standard 55-2020: Thermal environmental conditions for human occupancy, Tech. rep., ASHRAE (2020).
- [87] J. van Hoof, Forty years of fanger’s model of thermal comfort: Comfort for all?, *Indoor Air* 18 (3) (2008) 182–201. doi:10.1111/j.1600-0668.2007.00516.x.
- [88] P. Fanger, Thermal Comfort: Analysis and Applications in Environmental Engineering, Danish Technical Press, Copenhagen, 1970.
- [89] World Health Organization, Environmental noise guidelines for the european region (2018).
- [90] C. Backman, M. Hoeschele, Validation of a hot water distribution model using laboratory and field data, Tech. rep., National Renewable Energy Laboratory, DOE/GO-102013-39. (2013). doi:10.2172/1220208.
- [91] D. Zhang, K.-w. mui, L.-T. Wong, Establishing the relationship between occupants’ thermal behavior and energy consumption during showering, *Buildings* 13 (5) (2023) 1300. doi:10.3390/buildings13051300.
- [92] R. M. Company, R. tank water heaters product specifications (electric models 40-55 gal), 2025, Tech. rep., Rheem Manufacturing Company and Residential tank water heaters – product specifications (electric models 40–55 gal) (2025).
- [93] M. Shanahan, K. McDonell, L. Reynolds, Role play with large language models, *Nature* 623 (7987) (2023) 493–498. doi:10.1038/s41586-023-06647-8.

- [94] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: *Advances in Neural Information Processing Systems (NeurIPS)* 35, 2022.
- [95] V. Galaz, et al., Machine intelligence, systemic risks, and sustainability, beijer Discussion Paper 274 (2021).
- [96] T. Khot, H. Trivedi, M. Finlayson, Y. Fu, K. Richardson, P. Clark, A. Sabharwal, Decomposed prompting: A modular approach for solving complex tasks, in: *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.

Symbol	Definition
τ	Kendall’s rank correlation coefficient
ρ	Spearman’s rank correlation coefficient
w_i	weight of criterion i ($i = 1 \dots 4$), summing to 1
$v_i(x)$	value function score for criterion i at raw value x , range $[0, 10]$
α	shape parameter for logarithmic value functions
s_j	MAVT weighted sum score for alternative j
MAE	Mean Absolute Error on MAVT score (0–10 scale)
RMSE	Root Mean Squared Error on MAVT score (0–10 scale)
n	number of test scenarios evaluated
k	number of retrieved chunks per query in RAG ($k = 3$)

Table 7: Notation used throughout this paper.

Criterion	Weight (%)	Value Function	α	Direction
Environmental Impact	35	Linear	—	Lower raw = better
Energy Cost	30	Linear	—	Lower raw = better
Comfort	20	Logarithmic	1.5	Higher raw = better
Practicality	15	Logarithmic	1.2	Higher raw = better

Table 8: MAVT criterion specification. Weights sum to 100% and are held constant across all architectures and the ground truth calculator.

$$v_i(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \times 10 \quad (23)$$

$$v_i(x) = \frac{\ln\left(1 + \alpha \cdot \frac{x - x_{\min}}{x_{\max} - x_{\min}}\right)}{\ln(1 + \alpha)} \times 10 \quad (24)$$

				Decision Type	GT Calculator Receives	Withheld / Generalized for AI
Decision Type	Test Set (n)	RAG Set (n)	Total (n)	HVAC	R-value, SEER, HVAC age, sq. ft., insulation, occupancy context	R-value, HVAC age, SEER, occupancy context (insulation: good/avg/poor)
HVAC	60	25	85	Appliance	kWh/cycle, peak rate, off-peak rate, baseline time, appliance type	All five: type, rates, kWh/cycle, baseline time
Appliance	25	20	45		Shower	Tank size, water heater temp, GPM
Shower	15	5	20			
Total	100	50	150			

Table 9: Left: scenario distribution (BLOCKED in this draft; numbers stale – see audit log). Right: parameters available to the ground-truth calculator but withheld or generalized for AI input.

Metric	Pure Prompting	RAG-Enhanced	Hybrid
Ranking Accuracy			
Kendall’s τ	-0.06	0.43	0.80
Spearman’s ρ	-0.055	0.49	0.82
Top-1 Accuracy (%)	30.00%	58.00%	83.00%
Top-2 Accuracy (%)	62.00%	89.00%	97.00%
Score Error			
Overall MAE	2.52	1.61	0.69
Overall RMSE	3.07	2.40	1.52

Numbers BLOCKED in this draft; see audit log.

Table 10: Overall performance comparison (BLOCKED – stale numbers).

Criterion	Pure Prompting	RAG-Enhanced	Hybrid
Energy Cost	2.60	1.60	0.96
Environmental Impact	2.80	1.59	1.02
Comfort	2.51	1.97	0.32
Practicality	2.16	1.29	0.43
Overall	2.52	1.61	0.69

Numbers BLOCKED in this draft. Environmental MAE for Shower reflects water-consumption proxy (gallons); see Section 2.3.

Table 11: MAE per criterion by architecture (BLOCKED – stale numbers).

Decision Type	Pure Prompting			RAG-Enhanced			Hybrid		
	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)
HVAC	-0.11	25.00%	56.67%	0.52	63.33%	93.33%	0.90	91.67%	100.00%
Appliance	0.09	40.00%	72.00%	0.41	56.00%	88.00%	0.68	72.00%	92.00%
Shower	-0.11	33.33%	66.67%	0.11	40.00%	73.33%	0.60	66.67%	93.33%
Overall	-0.06	30.00%	62.00%	0.43	58.00%	89.00%	0.80	83.00%	97.00%

Table 12: Ranking accuracy by decision type (BLOCKED – stale numbers).

Metric	Pure Prompting	RAG-Enhanced	Hybrid
Total API Calls	300	300	100
Input Tokens	114739	295174	81741
Output Tokens	9970	13463	16275
Total Tokens	124709	308637	98016
Total Latency (s)	437.1	480.8	275.6
Avg. Latency/Call (ms)	1456.9	1602.6	2755.9
Success Rate (%)	100.00%	100.00%	100.00%
Failure Rate (%)	0.00%	0.00%	0.00%

Table 13: API usage and efficiency (BLOCKED – stale numbers).

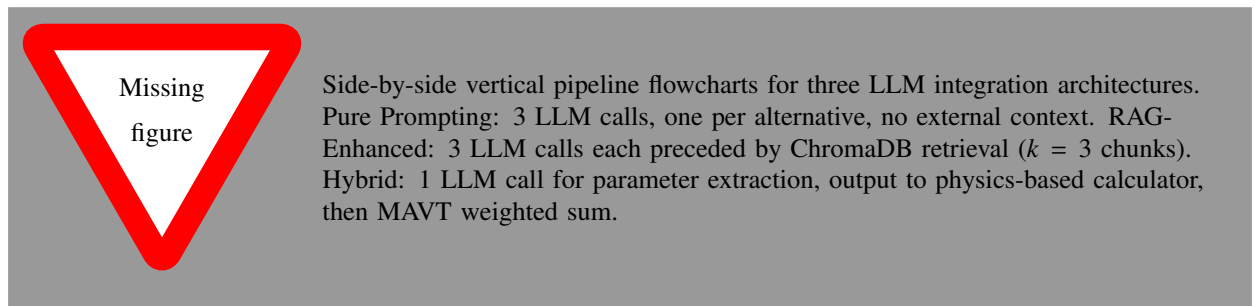


Figure 4: Processing pipeline for each architecture. Pure Prompting and RAG-Enhanced require three API calls per scenario; Hybrid requires one.